

Statement of Purpose

MS by Research in Data Science and Artificial Intelligence - CMInDS, IIT Bombay

Every year, millions of osteoporosis patients sustain life-altering fractures that could have been prevented with earlier diagnosis; yet the gold-standard diagnostic tool exists at a density of roughly 0.26 machines per million people in India. This gap between need and accessible technology is what first drew me toward research. What keeps me here is a question that sits at the heart of data science and AI: how do we build learning systems that are not only accurate, but well-calibrated, interpretable, and reliable enough to be trusted when it actually matters? My name is Adhya Jain, and I am applying to the MS by Research program in Data Science and Artificial Intelligence at CMInDS, IIT Bombay.

Academic Foundation

My undergraduate training in Computer Engineering at Mumbai University gave me grounding in Data Structures, Algorithms, Operating Systems, and Databases. Recognising early that coursework alone would leave gaps, I supplemented my studies through NPTEL particularly courses on Algorithms and Operating Systems which trained me to engage analytically rather than passively. I also pursued independent study in probability, statistics, and linear algebra as these underpinned the machine learning methods I was increasingly drawn toward. This habit of building foundations deliberately, rather than skipping to applications, continues to shape how I approach new problems.

Transition Toward Research

My practical work began with full-stack development, including early experiments integrating generative AI into applications. These projects were valuable, but more importantly, they shifted my attention from using AI models to interrogating them. I became curious about the learning process itself how models generalise, where they fail, and what it would mean for a model to be genuinely trustworthy rather than just highly accurate on a benchmark. I began studying machine learning through hands-on experimentation and project-building, which forced me to understand the statistical underpinnings of what I was building, not just the engineering surface.

Key Research Project: AI-Based Osteoporosis Detection

The most meaningful outcome of this exploration was the development of a deep learning system for early osteoporosis screening using standard X-ray images. The clinical motivation was direct: DEXA scans are expensive, scarce, and inaccessible to large portions of India's population, whereas X-ray machines are widely available. If a reliable screening method could be built from standard X-rays, early diagnosis could reach far more patients.

I designed a custom CNN trained on annotated knee X-rays, using augmentation techniques including random rotation, horizontal flipping, brightness jitter, and zoom variation to address class imbalance a problem with direct statistical consequences for sensitivity and specificity. A baseline single-model achieved 90% accuracy; an ensemble of five CNNs with soft voting improved this to 94%. Evaluation prioritised minimising false negatives, since the asymmetric cost structure of clinical decision-making demands it.

I integrated Grad-CAM++ to generate saliency maps highlighting bone regions most predictive of density loss, paired with an Ollama-based language model that produced interpretive summaries from these maps. This explainability layer was validated by Dr. Vijay Y. Patil, an orthopaedic specialist at Bombay Clinic, who confirmed that the model's attention aligned with clinical judgment. That validation made something concrete to me: a model's statistical performance and its interpretability are not independent properties in high-stakes settings, the latter is what determines whether the former can be used at all.

Research Interests

This project sharpened my research direction, but also broadened it. The problem I kept coming back to was not specific to medical imaging it shows up anywhere AI is used in the real world. A model can be accurate on paper but still fail in ways that are hard to catch, and when that happens in a consequential setting, the cost is real.

What I want to build, broadly, are systems that are not just accurate but genuinely useful systems that help people make better decisions rather than making decisions for them. That means a model should be clear about what it knows and where it is uncertain, so that the person using it can stay in control. It also means the system should work in messy, real-world conditions, not just clean datasets. I am interested in exploring these questions across different problem areas the specific domain matters less to me than whether the work has real impact on how people interact with and trust AI systems.

I want to build systems that help people make better decisions systems where the human remains in control and the model is a well-calibrated tool.

Why IIT Bombay - CMInDS DS&AI

CMInDS is precisely the environment I am looking for. The centre brings together faculty working on core questions in modern AI, including learning theory, probabilistic modelling, and robust, efficient learning, with a strong emphasis on both rigour and real-world applicability. The interdisciplinary structure of CMInDS matters to me: the problems I care about sit at the boundary of statistics, machine learning, and domain application, and a centre designed to work across those boundaries is where I want to develop as a researcher.

I am particularly drawn to work within the centre that focuses on uncertainty and reliability in learned models, as well as the opportunity to collaborate across both academic and industrial research settings. IIT Bombay's location within Mumbai's broader research and technology ecosystem further aligns with the applied orientation of my interests.

Goals

My long-term goal is to contribute to AI systems that can be safely and meaningfully deployed in consequential settings not as replacements for human expertise, but as tools that extend its reach and improve its reliability. Achieving that requires advances at the methodological level: in how we train, evaluate, and certify models. The MS by Research in Data Science and Artificial Intelligence at CMInDS, IIT Bombay is where I want to pursue that work.

I am eager to contribute to and grow within IIT Bombay's research community.

Sincerely,

Adhya Jain